

12.9.5 Richer notifications

Notifications are a key part of the Android user experience because they let applications show key updates and status information to users in real time. Android 3.0 plans to extend this capability, letting developers include richer content and control more properties. A new



builder class will let developers quickly create notifications that include large and small icons, a title, a priority flag, and any properties already available in previous versions. Notifications will offer more types of content by building on the expanded set of UI Widgets that are now available as remote Views.

12.9.6 Multiselect, clipboard, and drag-and-drop

The platform plans to offer convenient new interaction modes that developers can use. For managing collections of items in lists or grids, developers can offer a new multiselect mode that lets users choose multiple items for an action. Developers can also use a new system-wide Clipboard to let users easily copy any type of data into and out of their applications. To make it easier for users to manage and organize files, developers can add drag-and-drop interaction through a DragEvent framework.

12.10 High-performance 2D and 3D graphics

12.10.1 New animation framework

The platform includes a flexible new animation framework that lets developers easily animate the properties of UI elements such as Views, Widgets, Fragments, Drawables, or any arbitrary object. Animations can create fades or movement between states, loop an animated image or an existing animation, change colors, and much more. Adding animation to UI elements can add visual interest to an application and refine the user experience, to keep users engaged.

12.10.2 Hardware-accelerated 2D graphics